Y. Benjamin Perez C.

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Professional Summary

Final-year Computer Systems Engineering student (graduating Dec 2025) with 3+ years of hands-on experience in **embedded systems, FPGA design, and low-level programming**. Skilled in **Embedded C/C++, VHDL/Verilog, Python, and Java**, with practical knowledge of ARM Cortex-M, PIC microcontrollers, and FPGA platforms (Vivado, Quartus, NEXYS). Experienced across the full development lifecycle, including prototyping, debugging (oscilloscopes, logic analyzers), and documentation. Motivated to apply technical expertise to mission-critical embedded and defence technology projects.

Education

Brunel University of London Sep 2022 - Dec 2025
BEng (Hons) Computer Systems Engineering, Level 6 — Expected 2:2 *Uxbridge, UK*
- **Relevant Modules:** Embedded Systems, Digital Systems Design, Microcontroller Group Design Project, Computer Architecture and Interfacing, Autonomous Systems Design.

Waltham Forest College Sep 2021 - Jul 2022
Access to Higher Education Diploma: Computing and IT, Level 3 — 120 credits (Distinction) *London, UK*

Alpha Building Services Ltd. Sep 2020 - Jul 2021
City & Guilds Diploma: Electrical Installations in Building Structures, Level 2 — (Pass) *London, UK*

Experience

Heat Pipe and Thermal Management Research Group Apr 2024 - Jul 2024
Systems Design Engineer Intern *Uxbridge, UK*
- Designed and supported control panel architectures for heat pump and energy systems, applying structured system-level design practices.
- Prepared detailed **P&ID diagrams** and documentation to ensure operational safety and compliance.
- Assisted with lab-based prototyping, component integration, and equipment validation for energy management systems.

Professional Development Centre, Brunel University May 2023 - Jul 2023
Career Ambassador *Uxbridge, UK*
- Engaged with cross-functional teams to support university programmes and events, strengthening communication and organisational skills.

Complex Creatives (Digital Marketing Agency) Jun 2021 - Aug 2021
Junior WordPress Developer Intern *London, UK*
- Developed and deployed client websites using **WordPress** and **WPBakery**, with emphasis on reliability and secure configuration.
- Gained exposure to **eCommerce system security**, learning to apply plugin hardening and basic site vulnerability mitigation.

Projects

PIC16F18877 Temperature & Humidity Monitoring System Embedded C, Assembler, MPLAB X IDE
- Programmed PIC16F18877 microcontroller in **Embedded C/Assembly** to interface with LM35 and DHT11 sensors, LCD and 7-segment displays.
- Developed low-level drivers, tested sensor reliability, and validated outputs using breadboard prototypes.
- Documented design and verification process, aligning with structured engineering reporting.

Finite State Machine Line-Following Robot C, MSP432 ARM, Code Composer Studio, Logic Analyzer
- Implemented FSM-based control firmware in C on ARM MSP432 target for real-time robotic navigation.
- Used **Logic Analyzer** for real-time debugging and system optimization.
- Demonstrated deterministic firmware behaviour and embedded debugging methodology.

FPGA Sequence Detector System VHDL, NEXYS FPGA, Xilinx Vivado
- Designed and tested a **state-machine sequence detector** in VHDL with debouncing strategies.
- Simulated, validated, and performed timing/power analysis in Vivado.
- Experience transferable to low-latency, DSP, and hardware prototyping environments.

IoT Distance Monitoring System Embedded C++, Arduino, Python, ThingSpeak IoT

- Integrated motion sensors with Arduino, transmitting data to **ThingSpeak** for remote monitoring.
- Processed sensor data in Python and validated IoT pipeline reliability.

Tiny-ML Occupancy Monitoring on Edge Python, TensorFlow Lite, Raspberry Pi (Linux)

- Deployed **Conv1D CNN** on Raspberry Pi with Linux OS for real-time occupancy detection.
- Performed time-series analysis and implemented lightweight feature compression for efficiency.
- Experience demonstrates embedded Linux scripting and signal processing.

Technical Skills

Programming: Embedded C, C++, Python, VHDL, Verilog, Java, Assembly
Embedded Systems: ARM Cortex-M (MSP432, ESP32), PIC16F18877
Arduino, RTOS (exposure to FreeRTOS)
Tools/Platforms: Linux, MPLAB X IDE, Code Composer Studio, Arduino IDE
Xilinx Vivado, Quartus, NI LabVIEW
Debugging/Electronics: Logic Analyzers, Schematic Reading, Breadboard Prototyping
Signal Processing: MATLAB, Python (NumPy, SciPy, OpenCV)

Achievements & Leadership

Competitions: 1st Prize Winner, Brunel Blockchain Society Hackathon - **2024**
ICPC UK and Ireland Programming Contest - **2023**
Brunel R. E. A. D. Y. Programme - **2024**
Course/Student Representative Brunel University of London - **2025**
Mentoring Programmes: Graduate Aspirations Programme - **2025**
Brunel Professional Mentoring Scheme**2025**